DATA ANALYTICS

(Business Intelligence)

RUST

UNIT-1: Data Analysis Foundation

- → Univariate Analysis
- → Bivariate "
- → Multivariate "

- Graph Data
- Kernel Methods
- Working with High dimension data
- Dimensionality Reduction Methods

Friday
31 July, 2026

Data Analytics

- " " is the systematic computational analysis of data or statistics.
- It is used for the discovery, interpretation & communication of meaningful patterns in data.
- It is also entails applying data patterns towards effective decision-making
- Data Analytics is the science of examining, cleaning, transforming & modelling raw data to discover useful information, identify patterns, draw conclusions & support decision-making. In simple terms, it is the process of converting raw data into meaningful, actionable insights.

Histogram Padh lena

Types of data analytics

Lvl	Type	Question	Example
1	Descriptive Analytics	What happened?	EDA reports, Dashboards (Exploratory data Analysis)
2	Diagnostic "	Why did it happen?	Root cause analysis
3	Predictive "	What will happen?	ML Models, Forecasting
4	Prescriptive "	What should we do?	Optimization, Recommendation

* EDA report covers lvl 1 & 2

Key distinctions: Data Analytics vs Data Science

Aspect	Data Analytics	Data Science
Focus	Answering specific business questions	Discovering new insights & building models
Scope	Applied, problem-driven	Exploratory, discovery-driven
Output	Reports, dashboards, recommendations	Models, Algos, new questions
Skills	Statistics, SQL, visualization	ML, programming, advanced math

Monday
18 Aug, 2026

Dataset Analysis

62, 68, 72, 74, 77, 81, 90

(i) Mean: $\frac{524}{7} = 74.86$

(ii) Median: 74

(iii) Mode:

(iv) Standard deviation: $\sqrt{\frac{\Sigma(x_i - \bar{x})^2}{n}} = 8.75$

$= \sqrt{\frac{612}{7}} = 0.8285$

imp

Data Analysis Process

(1.) Define the process (ask)
Understand the business problem or research question, define objectives & success criteria. Ex - why sales are decreasing

(2.) Data Collection (prepare)
Gather relevant data from databases, CSV files, APIs, surveys, sensors or web scrapping; access data quality & availability

(3.) Data Cleaning (process)
Handle missing values, remove duplicate records, correct errors & standardize formats; often called data wrangling or data munging & typically the most time consuming step (60-80% of an analysts efforts)

(4.) Data Transformation
Convert data into a suitable format, normalization, importing categorical variables, feature scaling, aggregation

(5.) Exploratory Data Analysis (EDA)
Understand the data using histograms, pie charts, box plots, scatter plots & correlation matrices to spot patterns, detect anomalies & form hypotheses

(6.) Modeling (Analysis)
Apply techniques such as regression, classification, clustering or forecasting, depending on the business question

(7.) Evaluation
Assess the model or analysis using appropriate metrics (accuracy, rmse, silhouette score, etc.) & validate it against original business goal

(8.) Interpretation & Visualization
Interpret results in plain language (for sales increased due to festive season) & present findings via dashboards, charts & reports. Story telling with data is key

(9.) Act / Decision Making
Use the insights to make decisions, implement solutions or refine the process. This completes the loop: the process is cyclical not linear

Data Analysis follows a systematic, often iterative work flow. A widely referenced framework for this is CRISP-DM (CRoss Industry Standard Process for Data Mining).

H.W. Look for a research paper of ~~self~~ my interest. Find out dataset used in that paper. Look for the implementation of that paper.
Recent: 2024, 2025, 2026 Springer, Elsevier, ACM, IEEE
Journal

Tuesday 11 Aug, 2026

☆ Variance vs Standard Deviation

⇒ Dataset A
40, 45, 50, ~~56~~ 55, 60

⇒ Dataset B
10, 20, 50, 80, 90

Population Variance $= \Sigma(x - \text{mean})^2 / N$

Standard Deviation = $\sqrt{\text{Variance}}$

Measures of central tendency	Dataset A	Dataset B
Mean	~~45~~ 50	50
Mode	~~50~~ –	–
Median	50	50
Variance	50	1000
Standard Deviation	7.07	31.62

H.W. Plot histogram of both datasets.

Variance: It measures how far each value is from the mean
- Small variance → Data pts are close together
- Large " → " " " widely spread

Std Deviation: It is simply square root of variance
- Variance is measured in squared units, making it harder to interpret
- Standard deviation is measured in the same units as the original data, making it easier to understand
- Ex: Salary - ₹ ⇒ Variance - $(₹)^2$ & Std Dev. - (₹)

Variance	Std Dev.
• Avg. squared dist. from the mean	• Square root of variance
• Measured in squared units	• Measured in original units
• Harder to interpret	• Easier to interpret

#Variance & std. dev. are used in

- Exploratory data analysis
- Feature scaling
- Outlier detection
- Data distribution analysis
- Risk analysis
- ML algorithms

Real World Applications

- Finance - Measure stock market volatility
- Manufacturing - Check consistency in product quality
- Healthcare - Analyse variation in patient test results
- Machine Learning - Standardise features before training models.

#NOTE: Higher std. dev. simply means greater variability, not better or worse

Friday, 1 Aug, 2025

Types of Data Analysis

Univariate Analysis

- The simplest form of analysis, examining a single variable at a time to describe its distribution, central tendency, spread.

Aspect	Details
Common Measures	Mean, Median, Mode, Range, Std dev., Quartiles
Graphs Used	Histogram, Bar Chart, Pie Chart, Box plot, freq. table
Advantages	Easy to understand, summarize data quickly, detect outliers
Limitation	Can't identify relationships blw variables

- Example: Consider a list of Student Marks [70, 65, 85, 90, 75] Univariate analysis answers questions like- "what is the avg marks?" "what is the max/min?" & "How are the marks distributed?"

Bivariate Analysis

- Analysis involving exactly two variables simultaneously, to determine the Empirical relationship – association, correlation or dependency blw them.
- Ex: Hours studied v/s marks obtained
- Question answered, ~~thus~~ does studying more increase marks

Hours Studied	Marks Obtained
2	45
4	60
6	75
8	90

Variable	Typical Techniques
Numeric - Numeric	Scatter plots, Pearson/Spearman correlation, simple linear Regn.
Categorical - Categorical	Contingency tables (cross-tabulation), stacked bar charts, Chi-sq. tests
Categorical - Numeric	Side-by-side box plots, density plots, T-test / ANOVA

Advantages:

- Reveals relationships blw variables & supports simple prediction.
- For ex - Is there a relationship blw advertising spend & sales.

Multivariate Analysis

- Analysis involving 3 or more variables at the same time, capturing complex interactions & structure that ~~may~~ univariate or ~~not~~ bivariate analysis would miss, while also helping to control for confounding factors.
- Ex - predicting an employee's salary, using experience, educ. lvl, skill set & age together.
- No single one of these variables alone tells the full story

17/8/26

Data Analytics

Multivariate Analysis :-

Techniques	Application
Multiple lines / Logistic Regression	Medical Diagnosis, Financial forecasting
Principal Component Analysis (PCA), Factor Analysis	Image Processing, Survey data reduction
Cluster Analysis (K-means, Heirarchical	Marketing/Customer Segmentation
Decision Tree, Random Forest, SVM	Multifactor problems

→ Example:- Question answered:-

→ How can we group customers based on their purchasing behaviour across hundreds of products?

→ How do price, advertising, & competetor action combine to affect sales.

=> [Comparision] => Univariate, Bivariate & Multivariate

Feature	Univariate	Bivariate	Multivariate
No. of variables	one	Two	3 or more
Purpose	Describe	Relationship	Complex Relationship
Typical measure/ Technique	Mean, Median	correlation	PCA, Regression clustring
Typical visualization	Histogram	scatterplot	Heatmap, 3D plot, Parallel coordinates

Correlation = $\frac{\sigma_{xy}}{\sigma_x \sigma_y}$

σ_{xy} → Covariance ($Cov(x,y)$)

σ_x, σ_y → Standard deviation

Data Analytics : 18th August, 2026 : Tuesday.

High dimensional data:

Data describe by very large no. of features / attributes (columns), often large relative to the no. of observations (rows)/(tuples)

There is no strict cutoff, but the term implies that dimensionality itself is a significant factor in the analysis. For instance, a dataset with just 20 samples but 5000 features

Typical examples:

1. gene expression / microarray data: 1000's of genes measured on a few dozen samples.
2. Documents represented as vectors: over of thousand vectors on million of unique words.
3. Image data: Each pixel is treated as a feature. (even small image has 784 dimensions).
4. Sensor / IoT data: Many simultaneous reading per timestamp.

Characteristics and challenges:

Aspect	Impact
Sparsity →	Most values are often zero or missing possible words eg: a document uses only a fraction of all suitable
Multicollinearity →	Many features correlate with each other. Providing redundant information.
computational cost →	Many features mean needs more computation and storage.
overfitting risk →	Many features but comparitively few samples.
Interpretability & visualization →	Humans cannot directly visualize beyond 3 dimensions, making patterns hard to find.

Advantage of high dimension data:

1. Rich, detailed information content:
2. Better representation of genuinely complex problem.
3.

Disadvantages:

1. Increase computation Time.
2. Difficult to visualize or interpret.
3. Higher risk of overfitting.
4. Large Storage requirements.

Precisely

These challenges are what give rise to the curse of dimensionality and motivates the use of dimensionality reduction Techniques.

Monday
24 Aug, 2026

Curse of Dimensionality

→ A term coined by Richard Bellman describing a set of phenomena & challenges that arise when analysing & organizing data in high dimensional spaces, which simply do not occur in lower dimensional settings (2D-3D)

Key Effects:

1.) Data Sparsity
- To maintain the same density of data points as in lower dimensions, the amount of data needed grows exponentially with dimensionality.
- If 100 points are enough to densly cover 1 dimension, roughly 100^2 are needed for 2-dimensions, 100^3 for 3 & so on...
- High dimensional space ends up mostly empty

2.) Distance Concentration
- In high dimensions, the distance between the nearest & farthest data points becomes almost equal, that means all data points start looking equidistant
- This severely weakens distance-based methods such as k-NN, k-Means & DBSCAN

3.) Overfitting Risk
- With many features, there is a high chance of finding spurious correlation, models become flexible enough to fit the training data almost perfectly while generalising poorly to new data.

4.) Computational Complexity

- Time & memory requirements for many algos scale poorly, often quadratically or cubically with the no. of features.

5.) Need for Exponentially More Data

- To reliably estimate patterns at the same confidence level, the required no. of samples grows exponentially as dimensions increase

Solutions

1.) Feature Selection (filter/wrapper/embedded methods)
2.) " Extraction (PCA, LDA, Kernel PCA, Autoencoders)
3.) Removing Redundant or highly correlated features
4.) Collecting proportionally more data where feasible.

Dimentionality Reduction Techniques

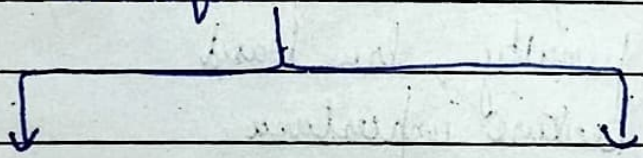
Techniques that reduce the no. of i/p features/variables under consideration while preserving as much meaningful info as possible, directly combatting the ~~CoDi~~. curse of dimensionality

Benefits

- Mitigates overfitting by removing irrelevant or redundant features
- Improves performance by reducing computational cost & training time.
- Enables visualization by projecting data into 2-D or 3-D.
- Improves data understanding by revealing the most imp. latent factors.

Neo4j

* Dimensionality Reduction Technique fall into 2 broad techniques



Feature Selection :- Selects a subset of the most relevant, original features, the remaining features are altered (For eg: choosing 10 useful features out of 100)

Method	How it work	Trade off
1) Filter Methods	Rank features using statistical measures independent of any model (correlation, chi square, info. gain)	Fast, but ignores how the model itself will use the features

Wrapper methods	Evaluate features subsets using a specific models performance (forward selection, backward elimination, recursive feature elimination	More accurate for that model, but computationally expensive
Embedded methods	feature selection happens during models training itself (Lasso regression's L1 penalty, tree based feature importance	Efficient middle ground between filter & wrapper method

★ Feature extraction → Linear Techniques
→ Nonlinear "

Transforms the original, high dimension space into a new smaller space

The new features are combinations (linear or non linear) of the original ones (for eg:- converting 100 features into 10 PC)

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Linear Techniques
→ PCA
→ LDA

NL techniques
→ Kernel PCA
→ t-SNE
→ UMAP
→ Auto Encoders (NN)
→ SVD

PCA

Unsupervised technique that projects data onto new, uncorrelated orthogonal axes (Principal components), chosen to maximise the variance captured [VC↑]
reduces redundancy & is widely used as a general purpose preprocessing & visualisation

LDA

Similar to PCA but supervised, it use class labels to find ~~to~~ the axes that best maximise separation b/w known classes, commonly used ahead of classification task.

90% → 1,70 → (40) → 90%.

40

Ep. 4

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Non-linear Technique

↳ Kernel PCA :- non linear extension of PCA that applies kernel trick to capture non-linear structure thereby doing dimensionality reduction.

↳ t-SNE :- t-distributed stochastic Neighbour Embedding primarily use for visualization, excels preserving local structure when projecting into 2D/3D, though computational heavy and stochastic results very run to run.

↳ Umap :- uniform manifold approximation & Projection
A more modern, generally faster alternative to tsne that aims to preserve both local & global data structure useful for both visualization & clustering

↳ Auto Encoders :- Neural-networks to reconstruct their own input through narrow bottleneck with hidden Layer.
- the compressed bottleneck serves as the reduce dimensionality features.
- Capable of capturing non-linear relationship.

↳ SVD :- Singular value decomposition
A matrix factorisation technique underline PCA, widely used in text analytics, recommender systems

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Comparison of techniques

Technique	Main purpose	
PCA	Maximize captured variance	Linear
LDA	Maximize class separation	Linear
Kernel PCA	Capture Non-linear structure	Non-linear
t-SNE	Data visualization	"
Umap	visualization & clustering	"
Autoencoder	Deep, learned feature compression	"

* Importance of Dimensionality Reduction Techniques in Data Analytics

- Speed up dimensionality analysis & modeling
- Reduces Storage & computation cost
- Removes noise & Redundant/correlated features.
- Enables visualization of otherwise high dimensional data

for examples :- Composing 50 Survey question into 2D scatter plot.

- helps mitigate overfitting by improving feature to Sample ratio.

D. {Eggs}

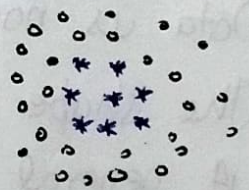
Tuesday
8 Sep, 2016

Q) What is a kernel function?

It is a mathematical function used to calculate similarity btw 2 data points without explicitly transforming the data into a higher dimensional space. In simple words, a kernel function tells us how similar two data points are.

Q) Why do we need kernel functions?

In many real-world datasets, the classes cannot be separated using a straight line. for ex- suppose one class is located inside a circle & other outside.



A straight line cannot effectively separate the two classes.

K.F. allow algorithms such as SVMs to handle such non-linear data

Basic idea: Original data
↓
Kernel Transformation
↓
Higher Dimensional feature space
↓
Linear Separation

Thus, K.F help SVM find a suitable non-linear decision boundary in the original feature space.

Common Kernel functions

1.) Linear Kernel : $K(x, x') = x^T x'$

2.) Polynomial kernel : $K(x, x') = (x^T x' + r)^d$

3.) RBF : $K(x, x') = e^{-\gamma \|x - x'\|^2}$

4.) Sigmoid Kernel : $K(x, x') = \tanh(\gamma x^T x' + r)$

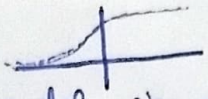
Choosing a kernel

1.) Use Linear Kernel when
(i) Data is approximately linearly separable
(ii) The no. of features is very large
(iii) Fast training is important
(iv) Text classification is being performed

2.) Use RBF Kernel when
(i) Data is non-linear
(ii) The shape of the decision boundary is unknown
(iii) A general purpose non-linear kernel is req.

3.) Use polynomial kernel when
(i) Polynomial relationships are expected
(ii) Feature interactions are important
(iii) The degree of the relationship can be controlled

4.) Use sigmoid kernel when



(i) There is a specific reason to model sigmoid like relationships.
(ii) Neural-nlw like behaviour is desired